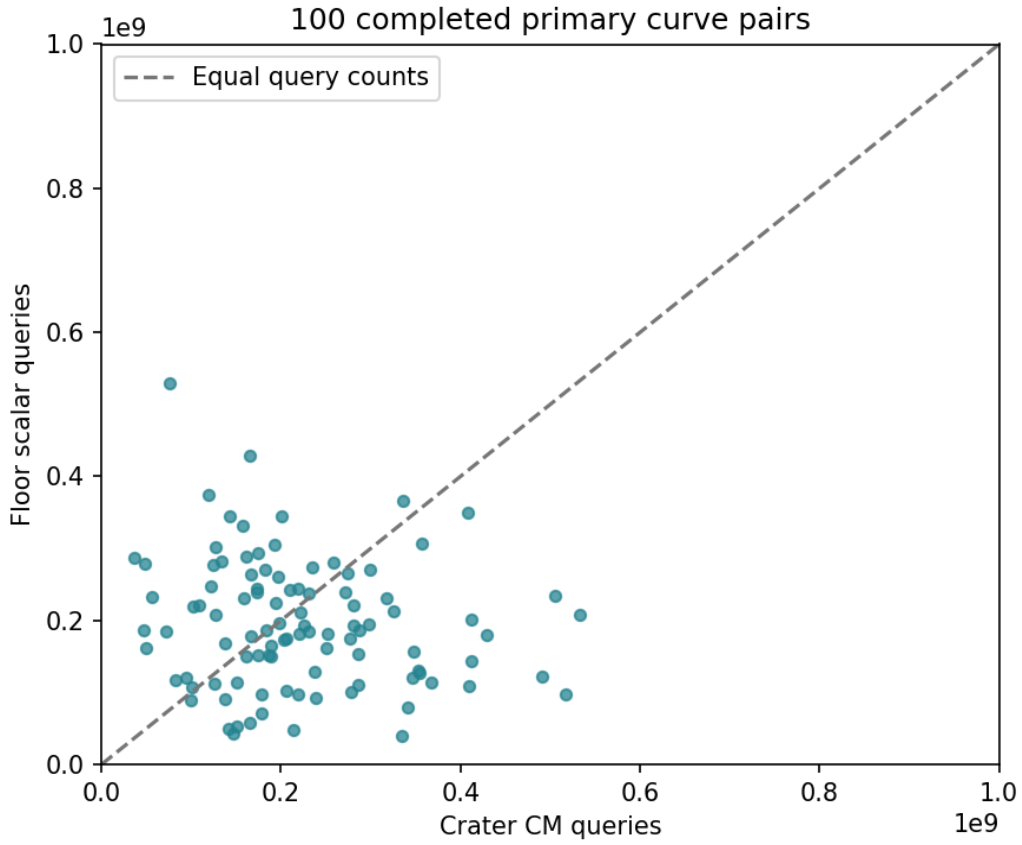
Conductor-gap research: completed native campaigns

15 October 2026 · Exact recoveries, structural certificates and remaining work

No systematic crater/floor mean timing difference is resolved in the 100-pair D = −19 population. A cheap CM endomorphism is concrete, while a generic recurrence gives the floor most of the same orbit-construction benefit. Intrinsic ECDLP hardness remains unestablished.

Completed experiment	Verified attempts / result
100-pair primary campaign	500 / 500: 300 folded IC and 200 same-point rho
Replication in 30 sessions	265 / 265: 155 IC and 110 rho
Same-crater map/scalar comparisons	145 exact agreements: 100 primary plus 45 replication
Independent replay	473 IC base/relation/matrix replays; 797 PARI target recoveries
Orbit construction	44,000 trials, including A/A, recurrence, pinned repeats and fresh-process batches

Primary floor/crater native-backend CPU: 0.980, paired 95% interval [0.866, 1.111]. Query ratio: 0.878 [0.764, 1.009]. Ten thousand whole-gap bootstrap resamples; all 100 endpoint pairs included.

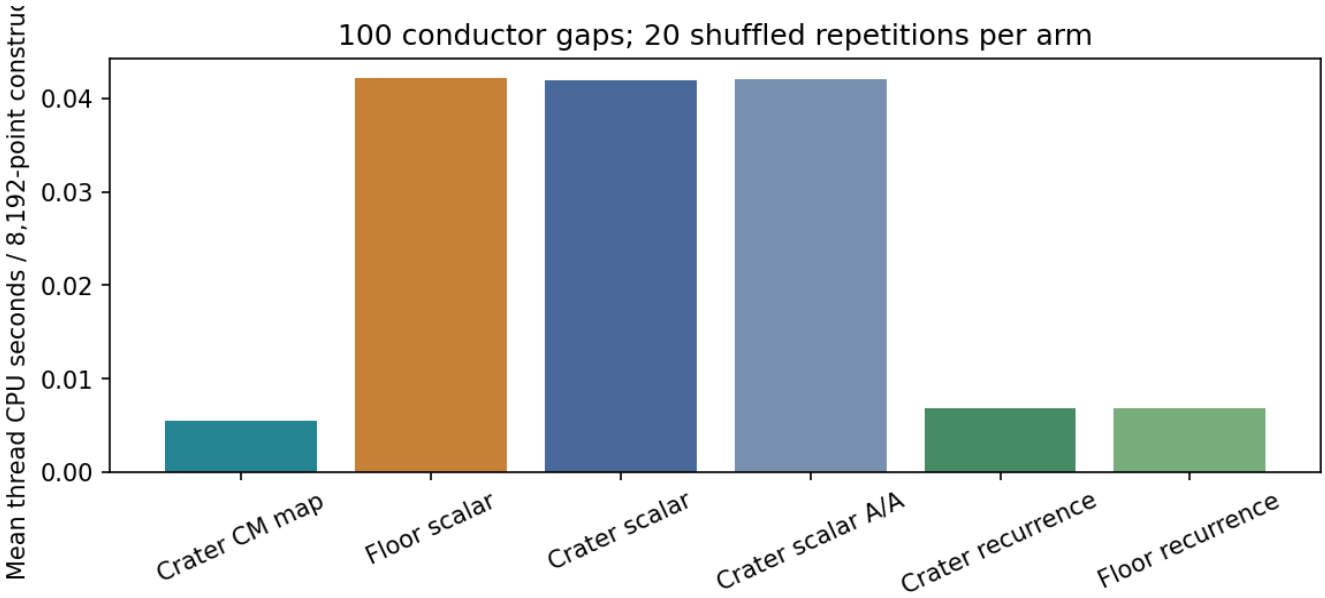


CPU observations are L0 diagnostics; endpoint strata ran mostly in separate queue waves. Bootstrap intervals do not remove host/wave confounding. Native-backend CPU includes orbit/table setup and native recovery. The measured process also includes parameter derivation and independent replay. Native prices remain incomplete; no complete-price IC/rho ratio or S=0 result is admitted.

1. The structural effect and its generic control

The tested pair family has $p = 5\ell^2 - \ell + 1$, $t = \ell - 2$, prime $N = 5\ell^2 - 2\ell + 4$, $D_K = -19$ and endpoint conductors 1 and ℓ. The 100 distinct prime gaps span 33,555,419–35,282,963; completed ECDLP comparisons use 53-bit fields.

On the crater a rational degree-5 CM map acts as [λ], where $\lambda^2 - \lambda + 5 = 0$ modulo N. The floor order has no degree-5 endomorphism. Nevertheless $P[k+2] = P[k+1] - [5]P[k]$, after one initial [λ] multiplication per seed, generates the same linked orbit with generic arithmetic on either endpoint.



CPU 0 pinned; local monitor uncontended. Physical-host/frequency isolation unverified.

Pinned means per 8,192-point construction: crater CM 5.464 ms; crater recurrence 6.859 ms; floor recurrence 6.857 ms; direct floor scalar multiplication 42.141 ms. The local monitor reported no contention: 0.34 other CPU seconds during 300.43 wall seconds. Physical-host and frequency isolation are unverified.

Twenty fresh-process batches add 12,000 construction records. Each batch contains all 600 model/arm jobs and uses precomputed CM parameters; it is not a separate process per arm. These batches ran at L0. First observed passes and later shuffled passes are recorded separately. Cold/warm cohorts are not a causal timing comparison.

Four public seed logs and 2,048 levels yield 8,192 factor-base points. Signed pair sums produce equations in four seed logs and the target. Scalar-proportional column parts are rejected; the matrix may pin the target before determining every seed log. Independent replay checks the base, each relation, rank and $[d]G = Q$.

For 137 cases, including the new 96-bit pair, independent PARI point-order and Weil-pairing witnesses certify crater-twist invariant factors ℓ and $q\ell$, versus a cyclic floor twist of order $q\ell^2$ ($q = 5$ or 11). This gives derived rational degree-ℓ kernel-line counts $\ell+1$ versus one. These counts are not a complete isogeny enumeration. Both original ECDLP endpoints retain the same prime order N.

2. Replication and the current curve

Cohort / gap clusters	Query ratio, floor/crater	Native CPU ratio, floor/crater
fresh / 10	0.988 [0.845, 1.149]	1.059 [0.912, 1.221]
isomorphic / 10	1.397 [0.882, 2.056]	1.465 [0.974, 2.049]
d43 / 10	1.375 [0.998, 2.011]	1.339 [1.006, 1.887]
pilot-five-rounds / 1	0.977 (descriptive)	0.951 (descriptive)

Intervals use 10,000 whole-gap cluster resamples, retaining all targets/rounds within each sampled gap. The five-round pilot has only one gap and therefore no curve-population interval. Twenty additional floor vertices completed another 40 IC/rho attempts.

The D = −43 native CPU interval narrowly excludes one in this ten-pair exploratory subset; the excess CPU accompanies extra queries. Intervals are unadjusted for multiple comparisons, and timings are L0. This is a replication lead, not a general conductor-hardness conclusion. The coordinate control transports G and Q but reruns prefix seed selection, so it also changes geometric seed points.

The completed 200-target rho panel has a posthoc floor/crater full-process CPU ratio 1.174 [0.99897, 1.38318], with mean CPU 14.335s / 16.834s. Its charged group-work lower-bound ratio is 1.160 [0.986, 1.367]. Both intervals include one. Canonicalisation remains unpriced; endpoint waves confound CPU. These within-rho diagnostics are not an admitted IC/rho speed comparison.

Pair 46 parameter	Exact value
p	5947698030651343
N = r; cofactor	5947697996161643; 1
t; prime gap ℓ	34489701; 34489703
Crater $y^2 = x^3 + ax + b$	a = 5757116383569871; b = 3560173328417442
Floor $y^2 = x^3 + ax + b$	a = 2087370704375809; b = 1402887937802154

The generic-recurrence pilot recovered both pair-46 targets and reproduced the direct-scalar factor-base hashes and query counts exactly. Across all completed sessions, 797 target recoveries passed independent PARI replay, including prime/Hasse witnesses and rejected scalar mutations. All 473 IC transcripts passed native public-base, relation and rank replay, including corrupted-sign rejection.

Six initial small-control preflight failures exposed loss of public fixtures in measured-child normalization. CurveSpec::for_child fixes that transport. Failed receipts remain preserved. The full Rust library run passed 4,116 tests (117 ignored); current fixture and CM-map checks and pinned Rust 1.98 strict Clippy also pass. Conditional Lean recurrence/orbit/norm lemmas are axiom-free; CM classification and point-map bridges are not claimed as formalized.

3. A certified 96-bit crater/floor pair

The largest independently certified prime conductor ratio is now 125,000,000,302,579. PARI independently proves primality, nonsingularity, exact group order by a Hasse-unique point witness, and floor conductor. This extends the additional floor certificates from 15 through 91 bits to 16 through 96 bits. It is a construction result; all completed large-gap ECDLP timings remain 53-bit.

96-bit parameter	Exact value
p	781250003782236254577699536 27
N = r; cofactor	781250003782235004577696510 51; 1
t; prime gap ℓ	125000000302577; 125000000302579
Fundamental discriminant	-19
Crater $y^2 = x^3 + ax + b$	a = 730994155585718132938198396 48; b = 226912769129733337099565752 23
Floor $y^2 = x^3 + ax + b$	a = 533537857111087148857771499; b = 482826264428683387500264558 88
Chosen floor generator	(1, 298417558966906745472366799 90)

A separate PARI torsion certificate supplies two independent crater-twist ℓ-torsion points, a nontrivial Weil pairing, and a floor-twist point of order ℓ^2. Derived rational degree-ℓ kernel-line counts are 125000000302580 versus one. These counts do not enumerate all isogenies. Full point transport and dual maps remain uncertified.

The optimized constructor passes 4,096 coefficient comparisons and both 100-codomain reference campaigns. Its guarded 96-bit run completed in 2,697.60 process wall seconds, with maximum RSS 9,362,772 KiB, under 11 GiB and 5,400-second caps. No local compiler or solver overlapped the run; the OOM counter did not increase. Timing is L0, without physical-host isolation. Prior parameter/kernel search and independent certification are outside this interval.

Earlier 6 GiB attempts hit time/allocation caps; the first 11 GiB attempt hit its one-hour cap. A 14 GiB precomputed-tree attempt was OOM-killed when test compilation grew. Those failures are preserved alongside the successful run. Other prepared upper models remain individually unconstructed.

The older near-limit candidate, ℓ = 129149576694539 and p = 792281625117984032632351147 31, is separate. Its nonzero kernel point needs extension degree 2,226,716,839,561 (twist point degree 4,453,433,679,122). A dense 96-bit basis would need about 26.7 TB per field element. This is a prerequisite barrier for the current rational-point method; other kernel representations are not excluded. Its floor is unconstructed.

4. Next steps and evidence limits

The earlier depth-20 degree-2 volcano remains a separate completed export: 21 levels, 1,048,576 explicit chosen-trace models and all intermediate strata. It is a smooth conductor ratio traversed by small steps. The large-prime benchmark samples endpoints and 20 additional floors, not every vertex of every prime-degree volcano. Earlier prefix-IC censored results are preserved and are not converted into solve times.

• Repeat D = −43 on more fresh targets and gaps, with the generic degree-11 recurrence control; preserve failures and use curve-level clustering.

• Transport the four geometric seed points in coordinate controls, as well as G and Q, to isolate arithmetic from prefix-seed selection.

• Price all native components before a complete IC/rho work claim. Retain cold setup and all five exclusive online phases, including independent replay; earn isolation for primary wall-time comparisons.

• Certify fast point evaluation and dual maps on the new 96-bit pair, then transport G and Q and charge the transport explicitly. The older non-rational-kernel candidate needs a compatible kernel representation.

• A prime-field Semaev/F4/SAT complexity claim requires a separately implemented and independently checked backend. The completed folded MITM experiment makes no such claim.

Sources: REPORT.md, native source, frozen input/binary manifests, results/paired-primary.json, results/replication-metrics.json, results/isolated-summary.json, replay and PARI certificates, and per-session reports. The original six-page summary and evidence ledger remain in inputs/historical-evidence-summary.zip. Rebuild: python native/render_report.py (rendering only). PR: github.com/aburan28/crypto/pull/1888.